

Carlos Moreno Rojas

Biomedical & Electrical Engineer | R&D | Embedded Systems | Instrumentation

Santiago, Chile | camoro2002@uc.cl

EDUCATION

Pontificia Universidad Católica de Chile <i>M.Sc. Candidate in Biomedical Engineering – Biosensors / Biomedical Instrumentation</i>	<i>2025–2027 (expected)</i> Santiago, Chile
Pontificia Universidad Católica de Chile <i>Professional Degree: Ingeniero Civil Eléctrico (Electrical Engineering)</i>	<i>2025–2026</i> Santiago, Chile
Pontificia Universidad Católica de Chile <i>Bachelor's Degree in Engineering Sciences – Biomedical Engineering (Major and Minor)</i>	<i>2021–2024</i> Santiago, Chile

ENGINEERING AND R&D EXPERIENCE

Research Engineer / Master's Researcher – Biomedical Instrumentation <i>Pontificia Universidad Católica de Chile – Dr. Rolando Rebolledo</i>	<i>2025–Present</i> Santiago, Chile
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- Re-engineered an **ex vivo organ-perfusion platform** and designed a modular monitor/data logger for pressure, flow, temperature, and vascular resistance using clinical sensors, ESP32-based electronics, and acquisition software.
- Solved pressure-pulsation, bubble, oxygenation, thermal, and fluidic-integration problems through iterative prototyping; calibrated measurements against certified references using correlation, deviation, and RMSE.
- Developed a quantitative **NIR/ICG imaging** approach combining optical-source sizing, filters, camera characterization, RAW acquisition, and calibration.

Hardware/Firmware Engineering Intern – Candel Medical Company <i>Medical Devices – Transcranial Electrical Stimulation</i>	<i>Jan.–Mar.</i> Chile
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- Redesigned/debugged circuits and PCBs and developed firmware on **ESP32/STM32** for a transcranial-stimulation device progressing from prototype toward product integration.
- Implemented BLE, signal generation and synchronization, protocol/peripheral control, power optimization, and experimental validation for tDCS/tACS/tRNS functionality.

Electrical Area Leader – Cervo UC / CYBATHLON 2024 <i>Robotic Transfemoral Prosthesis</i>	<i>2024</i> Santiago, Chile / Zürich, Switzerland
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- Led electrical/electronic development of a **robotic transfemoral prosthesis**, integrating prototypes with control and mechanical subsystems for CYBATHLON 2024 at ETH Zürich.

FPGA-Based Universal Testing Machine Upgrade <i>Meldic Ltda.</i>	<i>2025–2026</i> Santiago, Chile
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- Developed an embedded processing solution using **Zybo Z10, Vivado, and VHDL** for force, displacement, deformation, and speed-control signals in mechanical-testing equipment.

SELECTED PROJECTS

U-Net Medical Image Segmentation (2025): developed a U-Net neural network for lung segmentation in CT/MRI under Dr. Marcelo Andia.

Organ Preservation / Supercooling – Borealis UC (2024): contributed to supercooling-based preservation research and subsequent electromechanical redesign.

TECHNICAL SKILLS

Embedded / Electronics: ESP32, STM32, Arduino, C/C++, BLE, serial protocols, analog/digital electronics, PCB debugging, ADCs, HX711, PT100/PT1000, pressure/flow sensors, calibration, DAQ.

FPGA / Software: Vivado, VHDL, Zybo Z10, Python, signal/image processing, U-Net/deep learning, data analysis, acquisition software.

Design / Biomedical: Fusion 360, CAD, 3D printing, electromechanical prototyping, medical-device development, organ perfusion, physiological monitoring, NIR/ICG imaging, biosensors.

LEADERSHIP AND ADDITIONAL EXPERIENCE

President, Biomedical Engineering Student Chapter UC (2024–2025) | **Webmaster, IEEE EMBS PUC Chile** (2024–Present)

Laboratory Teaching Assistant, Physics UC FIS154 (Apr.–Jun. 2024)

LANGUAGES

Spanish: Native English: B2 / ALTE 3 (TOEIC)